

The HR Pairing System — Master Guide

Cameron's correlation-driven MLB home run parlay framework · v2.0

Compiled June 10, 2026. Evidence base: June 9, 2026 slate audit; May 29–June 8 ten-day look-back (~360 HR); academic hot-hand and correlation literature; live June 10 slate construction.

How to Use This Document (instructions for future sessions)

This guide is the standing system. When Cameron provides a daily data drop — Kasper slate reads (Matchup / Zone Fit / KHR columns), ULX model output, Outlaw or other public boards, odds screenshots — apply this framework in order: **CAUSE** → **COUNT** → **BATS** → **PRICE** → **STRUCTURE**. Research current probables, weather, and lineups; grade every candidate against Part III signals; assign pairings using Part IV archetypes; build the Part V structures; deliver a final bet order with conviction tiers, the standard card architecture (2-leg RR / 3-leg RR / straight megas / bonus-bet longshot), and explicit pre-lock checks. Push back on any leg that violates a veto rule, even if every external source likes it.

Cameron's metric definitions (from the Kasper tool): **Matchup** = purely historical batter-vs-pitcher-profile data, no recent form. **Zone Fit** = overlap between the batter's hot zones and the pitcher's typical pitch locations. **KHR** = the primary blended metric: Matchup weighted heavily by recent form (hot/cold adjustment). KHR covers the hitter side well; it does NOT grade pitcher batted-ball profile, weather, park, bullpen, or price — the framework supplies those layers.

Part I — Is Pairing Luck, or Can It Be Studied?

Short answer: the **edge** is studyable; the **sequencing** is luck. Three layers of pairing are genuinely researchable, and one layer is irreducible variance. Confusing the two is how bettors either quit a working system or trust a broken one.

Layer 1 — Within-game correlation: REAL and measurable

Events inside one baseball game are not independent. Outcomes within a single game commonly correlate at 30–50%+, which is exactly why sportsbooks build Gaussian copulas and correlation matrices to price same-game parlays — and why SGP hold runs 15–25% versus 4–5% on straight bets. The book's own behavior is the proof: they wouldn't tax same-game combinations that heavily if the legs were coin flips. For HRs specifically, the shared causes are concrete physics and personnel: one pitcher's broken fastball serves the whole lineup; one wind reading lifts every fly ball; one stretched bullpen extends the damage window for both teams. The 'hitting is contagious' literature (PLOS One) found statistically significant improvement in teammates' hitting while a streaking teammate is active — team-level offense moves together.

Layer 2 — Hot-hand persistence: REAL but modest

The landmark Green & Zwiebel study (~2 million MLB at-bats, 2000–2011, controlling for hitter/pitcher ability, handedness, and park) found a player's last ~25 at-bats significantly predict the next one — and a hot hitter is roughly **30% more likely to homer** than a cold version of himself. Later replications (FanGraphs 2025 and others) agree the effect is real but small, and that 'currently hot' is the only reliable predictor of staying hot. Translation for KHR: form-weighting is academically defensible — a 30% relative lift turns a 20% HR night into 26%, which is the entire margin on a +300 price. But it's a thumb on the scale, not a cause. Streak + matchup = anchor. Streak alone = watchlist. Our own 12-day audit matched the literature: Goodman (6 HR days), Kurtz (5), Dinger (6 HR), PCA (5 HR) kept repeating, while team clusters (Athletics ×3 separate nights) recurred all week.

Layer 3 — Structure math: PURE skill, zero luck

Leg counts, round-robin geometry, stake allocation, bonus-bet deployment, and line shopping are deterministic math. Two bettors with identical picks can have materially different ROI purely from structure. This layer is the most controllable and the most neglected.

Layer 4 — Sequencing: pure luck, by design

Even a perfectly built 3-leg correlated card hits maybe 4–6% of nights. WHICH night it hits is unknowable. The skill expresses itself only in the long-run average; any single slate is noise. Judge the process (was the cause right? did the pitcher actually get hit hard?) — never the night.

Verdict: pairing is a studied discipline wearing a lottery ticket's clothes. Correlation is real (the books' own SGP tax proves it), form persistence is real but small (~30% relative HR lift), structure is pure math — and the residual is variance you budget for, not variance you outsmart.

Part II — The Math of Pairing

1. The fundamental equation

$$P(A \text{ and } B) = P(A) \times P(B) + \text{Cov}(A, B)$$

Books price cross-game parlay legs by simple multiplication — they assume $\text{Cov} = 0$, which is correct across games. INSIDE a game, $\text{Cov} > 0$ for two HR legs. Define the **lift factor** $\lambda = P(B \text{ hits} \mid A \text{ hit}) \div P(B)$. The true pair probability is $P(A) \times P(B) \times \lambda$. If the book pays independence odds while $\lambda > 1$, the pair is +EV even when each leg is individually fair. Estimated λ for HR pairings:

Pairing type	Shared cause	Est. lift λ	Note
Same lineup vs same starter	One pitcher's stuff/command that day	1.25–1.60	Strongest. June 9: Gasser & Melton each fed 4 different hitters; pairs cashed within the same inning 3× on one slate
Cross-stack (one bat each side)	Park, weather, umpire, bullpens, game script	1.10–1.30	Half-right insurance: either broken starter helps
Same team, different game (stack over a series)	Lineup contagion + persistent park	~1.05–1.10	Weak but nonzero (PLOS One contagion)
Different games, shared weather system	Regional heat/wind	~1.0–1.05	Mostly diversification, not correlation
Different games, no shared cause	None	1.00	Pure independence — value must come from leg EV

λ estimates are directional, derived from clustering audits (e.g., June 9: 50% of 36 HRs off 7 pitchers; 39% from 2 of 15 games) — refresh them with each month's audit. Empirical anchor from the 10-day window: among teams that hit ≥ 1 HR in a game, roughly 6 in 10 hit ≥ 2 — versus ~35–40% unconditionally. Conditioning on one teammate's HR meaningfully raises the odds the game is a launch environment.

2. The SGP tax — and where it isn't

Books know about λ . Their SGP engines shorten correlated pairs, often overcharging to 15–25% hold. The actionable asymmetries: (a) some books still price two anytime-HR legs in the same game at or near independence — those books are where same-game stacks live; (b) cross-game legs are NEVER correlation-priced, so multi-cause cards across games carry only standard parlay vig; (c) when a same-game pair is taxed, compare the SGP payout to the independence product $P(A) \times P(B)$ — if the haircut exceeds your estimated λ benefit, split the legs into separate cards instead. Rule of thumb: pay the tax only when $\lambda \geq$ the payout haircut.

3. Edge compounding (why parlays of +EV legs are rational)

A parlay multiplies edges, not just risks. If each leg carries edge e (true probability $\div$ implied probability), an n -leg parlay's EV multiplier is $(1+e)^n$ before correlation. Three legs at +8% edge each $\approx$ +26% EV on the treble; add a same-game pair at $\lambda = 1.3$ and the card's theoretical EV is large even after vig. The brutal flip side: legs at –5% edge compound to –14% on a treble. **Parlays amplify whatever you feed them.** This is why no leg gets on a card without beating its implied probability — the structure can't rescue bad inputs.

Implied probability anchors: +200=33.3% · +250=28.6% · +300=25.0% · +350=22.2% · +400=20.0% · +450=18.2% · +500=16.7% · +600=14.3% · +700=12.5% · +800=11.1%

4. Leg count & survival math (30% per-leg baseline)

Structure	P(hit) indep.	With pairing lift	Role
2-leg	9%	12–14%	Survival layer — cashes 'partial bunch' nights
3-leg	2.7%	4–6%	Sweet spot — leverage with RR cover
4-leg	0.8%	1.5–2%	Straight lottery or RR shell only

Structure	P(hit) indep.	With pairing lift	Role
5-leg	0.24%	~0.5–0.8%	Nuke — correlation belongs HERE (Coors pairs ride together)

Key asymmetry: on ROUND ROBINS, diversify causes (independent legs maximize the chance SOME combo cashes). On STRAIGHT tickets, concentrate correlation (λ raises the joint probability and the book doesn't charge for it cross-ticket). Same five names, two opposite assembly rules.

5. Round robin geometry

3 picks by 2s = 3 doubles · 3 by 2s + treble = 4 tickets (DEFAULT) · 4 by 2s = 6 doubles · 4 by 3s = 4 trebles · 5 by 3s = 10 trebles · 4 by 2s+3s = 10 tickets. Break-evens at ~+300 avg legs and equal stakes: 2-of-4 on a 4×2s ≈ wash; 3-of-5 on a 5×3s ≈ 4–8× the day; any single double ≈ 1–1.5× total daily outlay. Decide TOTAL daily risk first, divide across tickets — never size combos individually.

Part III — The Signal Stack (grading every candidate)

A. Pitcher fade criteria (2+ flags = fade target; each fade is one CAUSE)

Signal	Threshold
HR/9 season / road split	≥ 1.4 (road split counts double for road starts)
HR/FB%, last 30 days	≥ 15% — recent form > season line
Barrel% allowed	≥ 9% (sample-stable contact quality)
Hard-hit% allowed	≥ 42%
Fastball shape	≤ 93 mph and/or flat ride at high usage; cutter-heavy soft-tossers count
K rate	≤ 18% — pitch-to-contact = 3–5 extra balls in play = extra HR exposure
Fly-ball rate	≥ 40% (multiplies with FB hitters); elite GB% ≤ 30% FB = VETO the spot
Experience/leash	Rookie ≤ 3 MLB starts, or manager allows 18+ batters (3rd-time-through spike)
Bullpen behind him	30-day reliever xFIP bottom-10 extends the cause (~40% of PAs are vs relievers)

B. Hitter signals, in order of power

- **1. Air-pull rate** — pulled FB+LD as % of batted balls (strip popups). The funnel: ~4% of all batted balls are HRs → 14.5% of fly balls → **37% of pulled fly balls**; 65% of pulled barrels leave. 20%+ air-pull is elite.
- **2. Zone/pitch-type fit** — usage-weighted per-pitch ISO/barrel vs the pitcher's actual arsenal (Kasper's Zone Fit column approximates this). Fastball-crusher vs 55%+ four-seam guy = elite fit.
- **3. Platoon ISO** — LHB vs RHP is the premium lane (shorter RF in most parks). Use ISO split, never AVG split. Verify reverse-split pitchers. Platoon riders batting 5–6 = the lineup card confirming the edge.
- **4. Form / KHR** — 3+ HR days in trailing 7–10 = legitimate ~30% relative lift (Green-Zwiebel). Multiplier on a cause, never a cause itself.
- **5. Batting slot** — slots 1–2: guaranteed 1st-inning look + 4–5 PA (a third of June 9 HRs came in innings 1–2); 3–4: anchor slots; 5–6: cheap platoon riders; 7–9: avoid outside Coors.
- **6. Pull-side porch fit** — pull LHB wants short RF; pull RHB wants short LF. Park factor is directional.

C. Weather (quantified)

~3 ft carry per +10°F (≈ +1% HR prob per °F) · wind OUT 10+ mph = independent cause (5 mph tailwind ≈ +19 ft; Wrigley 10+ out → over 60%+ since 2005) · wind IN 10+ = veto the game · humidity: humid air is LESS dense, tiny positive · Coors ≈ +5% carry, already priced — pair it with a pitcher cause · confirm roof status.

D. Standing vetoes (amended after June 10 audit)

Ace-tier high-K starter in form (streak + ace = pass) · Form with no attached cause (watchlist only) · Unconfirmed lineup (no bet until the card posts) · Wind in 10+ mph. **AMENDED — elite GB pitcher is now a DOWNGRADE, not a veto:** June 10, Framber Valdez allowed 2 HRs (Buxton, Lewis) — elite bats with elite form beat batted-ball profiles often enough that 'GB pitcher' caps the leg at satellite tier rather than killing it. Veto survives only for non-elite bats vs GB aces.

Part IV — Pairing Archetypes (ranked by λ)

Archetype	Construction	When
A. Same-pitcher stack (λ 1.25–1.6)	Two bats, same lineup, vs one fade target. One from slots 1–3 + one from 4–6 to span both trips through the order	Whenever a 2+ flag fade target exists. Place at the book that doesn't correlation-tax HR pairs
B. Cross-stack (λ 1.1–1.3)	One bat each side of a projected slugfest (both starters flagged, high total)	Coors-type games; half-right insurance
C. Park/weather stack (λ ~1.0–1.05)	Best bat from each of the 2 best environments	Wind-out days, heat waves, Vegas/Coors doubleheaders of causes
D. Boost anchor + satellite	Boosted/discounted leg anchors; satellites share its cause	Any day boost inventory exists; bonus bets → max-odds longshots ONLY
E. Hidden double	Elite barrel + air-pull bat in an elite spot; anytime price carries multi-HR tail cushion	Anchor selection tiebreaker (multi-HR games happened EVERY day of the 10-day audit)

Cause counting sets leg count: 1 strong cause → 2-leg stack (or 3×2s on that game). 2 causes → 3-leg with RR cover. 3+ causes (full 15-game slates reliably offer them; short slates don't) → 4–5 name pool, full architecture. **If you can't say the pair's shared cause out loud, don't pair them.** Anti-pattern: the 'three stars' card — famous bats, three unrelated games, star-taxed prices, $\lambda = 1.0$.

Part V — Standard Card Architecture (the June 10 template)

From a graded board, take the top 4 as the CORE (each must carry its own cause), the best value leg as the FIFTH (trebles only is acceptable — but know a 2-hit night including the fifth leg can cash zero), and the best correlated environment bat as the SIXTH (nuke only).

Layer	Build	~Share of cash	Job
2-leg RR	Core 4 by 2s = 6 doubles	30–40%	Survival: any 2-of-4 ≈ wash or better
3-leg RR	Core 4 + fifth, by 3s = 10 trebles	35–50%	Leverage: any 3-of-5 = 4–8× day
4-leg straight	Core 4, no RR	5–10%	Mega: all causes fire
5-leg straight	Core 4 + correlated sixth (e.g., both Coors bats ride together)	3–7%	Nuke: correlation is FREE on straight tickets
Bonus-bet treble	3 'weird value' guys, +450 or longer, each with a real cause + 2-source convergence; lineup-confirmed	\$0 cash	Bonus bets pay profit only → max-odds is their mathematically correct home

Discipline rules: fixed daily allocation (≈1.5–2% of bankroll across ALL tickets) decided before the slate · doubles ≥ trebles share on thin slates · never chase after a miss · skip days are bets too (zero clear causes = no card) · check each book's void rules on the concentration game (a Coors postponement shouldn't be a surprise at grading) · profit boosts → the straight mega; bonus bets → the freak show.

Part VI — Daily Workflow

Step	Action
1. Slate scan	Rank games by HR environment: park, weather/wind, total, roof. Shortlist 3–4. Note slate size (15-gamer → full architecture; ≤9 games → lean 2-leg)
2. Fade ID	Research EVERY probable starter in shortlisted games vs Part III-A (web search current stats — rookies and recent call-ups are prime targets, e.g., Paredes/Sproat June 10). Flag 1–3 with 2+ criteria
3. Cause count	Count independent causes → set leg count BEFORE seeing prices
4. Bat selection	Grade hitters per cause: KHR + Zone Fit + air-pull + platoon ISO + slot 1–6. Require 2+ source agreement (Kasper / ULX / framework / public boards). Cross-source convergence on a DEEP name is a strong signal
5. Price check	Shop DK/Caesars/Fanatics. Each leg must beat its implied probability. Note boost/bonus inventory. Same-game pairs → the book without HR correlation pricing
6. Structure	Build Part V architecture. RRs = diversified causes; straights = concentrated correlation
7. Pre-lock check	Lineups posted (platoon + weird guys especially) · starters unchanged · weather/roof unchanged · say every pair's cause out loud
8. Log	Record legs, causes, prices vs close (CLV), structure, stake. Grade CAUSE separately from RESULT next day

Part VII — Audit Protocol (how the system learns)

- **Cause accuracy** (the real KPI): for each fade target, did he allow a HR / 2+ barrels / exit early? A night where the cause fired but your bat missed (teammate homered instead) is a process WIN. Target: causes firing $\geq 50\%$ of the time.
- **Leg hit rate vs implied**: monthly, compare actual HR rate of your legs to their average implied probability. Beating implied by 3–5 points = the system works regardless of recent P&L.
- **λ check**: when one stack leg hit, how often did its partner? Refresh the Part II lift table with real numbers.
- **Calibration**: bats rated ~30% by model should homer ~30% of the time; shade outputs if the top bucket underperforms.
- **CLV**: closing line value on every leg. Consistently beating the close = long-run edge, full stop.

Part VIII — The June 10 Post-Mortem (44 HR / 15 games): Five Amendments

A full-wipe night with strong process. The audit found the information was largely PRESENT in the combined data (~13 of 44 HR hitters appeared somewhere in Kasper/Outlaw/framework discussion — Happ, Wood, Eldridge, Alonso, Suárez, Bohm, Buxton, Walker, Velázquez, Alvarez, Mead, Stowers, Burleson) but the SELECTION layer concentrated on the wrong subset. The failures were structural, not informational:

Finding	Evidence (June 10)	Amendment
1. Right cause, wrong bat — the #1 leak	Sproat fired → Alike Williams (not Kurtz). Detmers fired → Whitcomb (not Altuve). Rodón fired → A. Martínez (not Hoskins). Ray fired → Wood (ranked #11, unbet). Kirby → Alonso (rated, unbet). FIVE causes fired; zero bet legs cashed	COVERAGE BEATS CONVICTION. A fired cause yields ~1 HR among 9 bats; one bat captures it ~20–30%. On strong causes, prefer 2 cheaper bats (mini-stack) over 1 premium bat, or a team-to-hit-HR market when fairly priced. $P(0\text{-for-5 fired causes at 25\% capture}) = 0.75$ ■ $\approx 24\%$ — full wipes are a MODAL outcome of single-bat selection
2. The loud fade didn't fire; quiet fades did	Lorenzen (8.01, Coors, everyone's anchor) and Paredes: zero HR allowed. Meanwhile 8 pitchers allowed 2+: Patrick, Lambert, Valdez, Peterson, R. Nelson, Scherzer, Schultz, King — none of them the consensus fade	PUBLIC-HEAT CHECK. When every source + the public is on one fade, the price is worst, the hook is quickest, and pitchers nibble. Prefer 2-flag pitchers NOBODY is discussing. The consensus fade still plays — but capped, never anchored alone
3. Bullpen nights exist	50% of HRs came in innings 7–9; ~45% off relievers. June 9 was a starter/1st-inning night (33% in inn. 1–2); June 10 was the opposite regime	BAD BULLPEN = STANDALONE CAUSE (promoted from extender). Prefer bats who play all 9 (no late platoon removal). The anytime-HR bet is 60% starter / 40% pen — weight both
4. Cause concentration	Coors carried the heaviest share of the slip (PCA + Goodman + Happ + Suzuki exposure) and went quiet vs starters	CAP ANY SINGLE GAME/CAUSE at ~33% of total daily cash. The slugfests were WSH@SF (7 HR) and CIN@SD (4) — causes we had ranked but underweighted vs the famous park
5. What WORKED — keep it	Velázquez (+470 deep-convergence pick) homered → bonus treble went 1/3 on free money. Happ cashed within the Lorenzen-game cause (off the pen). Eldridge, Wood, Suárez, Bohm, Alonso — all flagged, all hit. Spend stayed capped at \$183	Convergence-on-deep-names is real signal. Cause identification is working (5+ fired). Structure discipline held. The leak is bat-level capture — fix #1 and #4 and the same night cashes 2–4 doubles

The honest math of the wipe: with 4 core causes at ~45% fire rate and ~25% bat-capture given fire, each leg is ~11–15% true — and $P(\text{zero doubles from the 6-double RR}) \approx 50\text{--}60\%$ on any given night. One blank slate is not evidence against the system; thirty days of cause-fire rate below 40% would be. Grade the month, not the night.

Quick-Reference Card (v2.1)

CAUSE → COVER → BATS → PRICE → STRUCTURE. Find broken pitchers (2+ fade flags) — prefer QUIET fades over consensus ones — plus bad bullpens and boom environments. Causes set legs: 1→2, 2→3, 3+→full architecture. COVER each strong cause with 2 bats (mini-stack) before adding new causes; cap any single game at ~33% of daily cash. Bats: air-pull + zone fit + platoon ISO + KHR form + slots 1–6 + plays-all-9, lineup-confirmed, 2+ sources. Every leg beats implied; shop 3 books; boosts→mega, bonus bets→max-odds deep-convergence treble. RRs diversify causes; straights concentrate correlation. Fixed daily risk. Grade the cause-fire rate and bat-capture rate monthly — never the night.

Evidence appendix available in companion doc: [MLB_HR_Pairing_Framework.pdf](#) (June 9 case study tables, 10-day look-back detail, advanced signal stack).

Part IX - v2.2 Amendment: The Capture Layer

Compiled June 23, 2026. This appendix amends the standing system after the June 10 audit and formalizes the missing decision between finding a good cause and choosing the exact batter(s).

New standing order: CAUSE -> CAPTURE -> COVER -> BATS -> PRICE -> STRUCTURE

The existing framework was already correct to grade causes, require multi-source agreement, compare price against implied probability, and separate RR logic from straight-ticket correlation. The v2.2 upgrade adds a mandatory capture decision before selecting individual hitters.

1. Capture method comes before player selection

After identifying a pitcher, bullpen, park, or weather cause, choose the capture method first: single bat, 2-bat mini-stack, team-to-hit-HR market, correlated game pair, or pass. Strong causes deserve coverage. Weak causes deserve precision.

Cause Strength	Definition	Default Capture	Exposure Rule
A+	Bad starter + bad bullpen + HR park/weather + multiple hitter fits + 2+ source agreement.	2-bat mini-stack OR team-to-hit-HR market if fairly priced.	Can anchor, but cap one game/cause near 33% of daily cash.
A	Starter has 2-3 fade flags and at least two viable bats fit the pitch/zone/profile.	2-bat mini-stack if prices are usable; otherwise best bat + satellite.	Core-eligible. Prefer one slot 1-3 and one slot 4-6.
B	One clear pitcher weakness, neutral environment, or only one standout hitter profile.	Single best bat only.	RR/core only if price is strong and lineup is confirmed.
C	Mostly form, name value, or one public model mention with weak cause support.	Watchlist or bonus-only.	No core exposure.
Public mega-cause	Every source is on the same fade, odds are steamed, and the market is crowded.	Cap exposure; do not anchor alone.	Satellite only unless price still beats implied.

2. The capture formula

Cause EV only becomes bet EV after capture probability. A fired cause often produces one HR among nine bats. A single selected bat may capture only 20-30% of those fired causes. Mini-stacks increase capture, while RRs protect against the remaining miss rate.

Term	Meaning	Daily Use
Cause-fire rate	Did the pitcher/bullpen/environment actually give up HR-quality damage?	Audit separately from whether your selected bat hit.
Bat-capture rate	When the cause fired, did your chosen hitter get the HR instead of a teammate?	Primary leak metric. Below 25% means too much single-bat conviction.
Coverage cost	Extra stake or odds penalty required to cover a cause with two bats or a team HR market.	Accept it only on A/A+ causes.
Public heat	How crowded the fade/hitter is across public boards and shortened prices.	Downgrade one tier if heat is high and price is not compensating.

Part IX continued - Updated Scoring

3. 100-point leg score

Every hitter can now be graded on the same scale. The goal is not fake precision; it is consistency. Scores should be used for tiering, not as absolute HR probabilities.

Category	Pts	Inputs
Pitcher Cause	30	HR/9, HR/FB, barrel% allowed, hard-hit% allowed, K rate, fly-ball rate, arsenal shape, recent command, leash, rookie/3rd-time-through risk.
Hitter HR Profile	35	Pulled-air profile, barrel rate, blasts/squared-up contact, platoon ISO, pitch-type fit, zone fit, KHR/form.
Environment	15	Park HR factor by handedness/pull side, porch fit, temperature, wind, humidity, roof, altitude, game total.
Lineup/Opportunity	10	Confirmed lineup, batting slot, PA expectation, plays all 9, platoon/pinch-hit risk.
Market/Structure	10	Price vs implied probability, line movement/CLV, source convergence, whether the leg fits RR or straight correlation.

Score	Tier	Use
80+	Core	RR-eligible; can anchor a 4-core structure if lineup/weather are clean.
72-79	Satellite	Straight pair, 3-leg add-on, or 5th RR/treble leg.
65-71	Bonus/value	Only with +450 or longer, 2-source agreement, or as part of a mini-stack.
Below 65	Pass	No bet unless a major late price/boost changes the market layer.

4. Recommended starting weights

Signal	Weight	Rule
Pulled-air / air-pull	Highest hitter weight	If he does not lift and pull, require an overwhelming pitch/zone or park reason.
Barrel + blast quality	High	Use barrels for contact quality; use blasts/squared-up contact as a tiebreaker, not a standalone override.
Pitch/zone fit	High	Usage-weighted pitch fit beats generic season stats.
KHR/form	Medium	Multiplier on a cause, never a cause by itself.
Lineup slot	Medium	1-4 is premium; 5-6 is playable; 7-9 needs a big price or elite environment.
Public heat	Negative	Downgrade chalk when the price has already absorbed the edge.

5. Lineup/opportunity penalties

Condition	Adjustment
Confirmed top-4 lineup spot	+3
Confirmed 5-6 spot	+1
7-9 spot	-2 unless Coors/wind/cheap value
Likely pinch-hit or defensive replacement risk	-5
Lineup not confirmed	Hold. No lock until posted.

Part IX continued - Ticket Scoring and Audit Changes

6. Ticket score - separate from leg score

A good leg can still make a bad ticket. Score the final ticket so RR cards and straight tickets are judged by different jobs.

Ticket Variable	0	1	2	3
Cause diversity	Same fragile cause	Two causes	Three causes	Four+ clean causes
Correlation clarity	None	Loose same slate	Same environment	Clear same-pitcher/game stack
Price quality	Bad/steamed	Fair	Good	Great/boosted
Lineup/weather confidence	Unknown	Partial	Mostly confirmed	Clean/confirmed

Interpretation:

- RR tickets should score high on cause diversity, even if correlation clarity is low.
- Straight tickets should score high on correlation clarity, even if cause diversity is lower.
- If a 3-leg ticket is three unrelated stars with star-taxed prices, it is a ticket-score fail even when the individual bats look good.

7. Updated daily audit fields

Field	Log it as	Why it matters
Cause grade	A+, A, B, C, public heat	Shows whether you are finding real damage spots.
Capture method	Single, mini-stack, team HR, pair, pass	Forces the most important decision into the log.
Cause fired?	HR allowed, 2+ barrels, early hook, bullpen damage	Process KPI.
Bat captured?	Selected bat HR / teammate HR / no HR	Measures right-cause-wrong-bat leakage.
Ticket score	RR score + straight score	Prevents structure drift.
Closing line	Open, bet, close	CLV is the long-run market check.

8. One-line v2.2 rule

After identifying a cause, choose the capture method before choosing the player: single bat, 2-bat mini-stack, team HR market, correlated pair, or pass. Strong causes deserve coverage; weak causes deserve precision.

Part X - Tool Roadmap: Daily Slate Capture Engine

Goal: automate the research stack without outsourcing the judgment. The tool should surface causes, rank capture methods, score hitters, and output bet structures - not blindly print picks.

1. Recommended data stack

Need	Best Source	Use	Notes
Slate, game IDs, venues, probables	MLB Stats API / MLB.com	Daily schedule, gamePk, venue, probable pitchers, status, first pitch.	Public endpoints are widely used but not guaranteed stable; cache responses.
Lineups	MLB Stats API when posted; fallback manual/lineup sites	Confirm batting order, handedness, defensive/pinch-hit risk.	Do not lock props before lineup is confirmed.
Pitch-level Statcast	Baseball Savant CSV or pybaseball	Pitch arsenal, velocity, movement, zone/location, batter results, recent splits.	Use caching; large pulls should be batched.
Batted-ball and bat tracking	Baseball Savant	Barrel, hard-hit, launch/spray, pulled-air proxy, bat speed, blasts, squared-up.	Bat tracking is a tiebreaker; not a standalone override.
Park factors	Baseball Savant + Ballpark Pal	Baseline handedness park factors plus daily hitter-specific park/weather factor.	Ballpark Pal API/docs may require access; build a fallback.
Weather	Open-Meteo or NWS	Temperature, wind speed/direction, humidity, precipitation, roof flags.	Open-Meteo is easy/global; NWS is best for US gov open data.
Splits/projections	FanGraphs CSV exports / pybaseball	Platoon ISO, projections, team/bullpen quality.	Respect ToS and cache; use manual exports if needed.
Odds	The Odds API or manual sportsbook entry	Anytime HR prices, boosts, line movement, implied probability.	Free tiers are limited; sportsbook scraping is fragile and risky.

2. Minimum viable product

- Backend: Python + FastAPI for API routes and scoring jobs.
- Database: Postgres/Supabase for players, games, stat snapshots, lineups, scores, bets, and audit logs.
- Jobs: scheduled morning pull, lineup refresh every 10-15 minutes near lock, final weather refresh, post-game audit.
- Frontend: Streamlit for quickest MVP, or Next.js/React if you want a polished hosted dashboard.
- Hosting: Railway/Fly/Render for the app, Supabase for database, GitHub Actions or cron jobs for pulls.

Part X continued - Tool Logic and Screens

3. Daily pipeline

Step	Job	Output
1	Sync schedule, venues, probables, game status.	Game table with starter, venue, time, roof/manual flags.
2	Pull season + trailing 7/14/30-day pitcher and hitter data.	Pitcher fade flags and hitter HR profile scores.
3	Pull weather by venue coordinates and game time.	Temperature, wind vector, humidity, precip risk, carry score.
4	Merge lineups when posted.	Lineup/opportunity penalties, plays-all-9 risk, PA expectation.
5	Run cause scoring and capture decision.	Cause Cards: A+/A/B/C plus recommended capture method.
6	Run hitter scoring inside each cause.	Hitter Cards: leg score, source convergence, price threshold.
7	Ingest odds manually or via API.	Implied probability, value gap, CLV tracker.
8	Build card structures.	RR core, same-game straights, bonus longshot treble, pre-lock checklist.
9	Post-game audit.	Cause fire, bat capture, ticket score, monthly calibration.

4. Dashboard screens

Screen	What it shows
Slate Board	All games ranked by HR environment, weather, total, and cause count.
Cause Cards	Each pitcher/bulpen/game environment with fade flags, public heat, and recommended capture method.
Hitter Cards	Top bats within each cause with leg score breakdown, price, lineup status, and source overlap.
Pair Builder	RR candidates, same-game stack candidates, cross-stack candidates, and payout calculator.
Pre-Lock Monitor	Lineup confirmed, starter unchanged, weather/roof unchanged, price still playable.
Audit Log	Cause fired, bat captured, CLV, structure result, monthly hit-rate calibration.

5. Build priority

Phase 1 should avoid the hardest problem: automated odds/player-prop collection. Start with MLB schedule, Statcast/Savant, weather, manual odds entry, and the scoring engine. Once the scoring output is useful, add odds APIs and automatic line shopping.

Phase 1 MVP: Slate Board + Cause Cards + Hitter Cards + manual odds + card builder. Phase 2: automated odds, Ballpark Pal API/fallback model, and post-game audit automation. Phase 3: backtesting and model calibration.

Part XI - Source Notes and Compliance

Use open/public data carefully. Cache, rate-limit, and keep source attribution. The goal is personal research automation, not reselling raw feeds.

Source	What to trust it for	Caution
MLB.com / MLB Stats API	Official slate, game status, probable pitchers, venue IDs, team/player IDs.	Public API wrappers warn that endpoints can change and are not formal official documentation.
Baseball Savant	Statcast, park factors, pitch movement, batted-ball data, bat-tracking definitions.	Large CSV pulls should be limited, cached, and batched.
FanGraphs	Exports for leaderboards, splits, projections, and contextual stat checks.	Use CSV export or allowed access; avoid aggressive scraping.
Ballpark Pal	Daily game and hitter-level park/weather context.	API/docs access may be gated; use as premium layer or build a fallback from Savant + weather.
Open-Meteo	No-key JSON weather, hourly forecasts, multiple locations, historical data.	Free non-commercial tier has daily limits and attribution requirements.
NWS	US open weather data, alerts, hourly forecast via gridpoint.	Requires geocoding and reasonable rate-limit behavior.
Odds APIs	Market prices, line movement, book comparison.	Free tiers are small; player HR props coverage varies by provider.

Implementation guardrails

- Separate source snapshots from model outputs so audits can reproduce what the tool knew at lock.
- Never overwrite a game snapshot after lock; create a new versioned snapshot.
- Tag each leg with the data timestamp used for schedule, lineup, weather, odds, and scoring.
- A tool-generated card must still pass the human pre-lock checklist: lineup, starter, weather/roof, price, and pair cause spoken out loud.
- Do not chase after model output. The tool should lower decision fatigue, not increase bet volume.

Final v2.2 mantra: Automate the research. Mechanize the scoring. Human-check the context. Structure the risk.